# Linear Algebra and Its Applications

## Section 2.9: Dimension and Rank

Detailed worked solutions

Mathjock

**Text and numbering.** David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

**Source scope.** The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

**How to read the solutions.** Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

## Contents

**2.9 Dimension and rank** . . . . . 3

Practice problem 2.9.P1: Count independent directions rather than generators . . . . . 3

Practice problem 2.9.P2: Turn coordinates into a vector . . . . . 3

Practice problem 2.9.P3: A subspace cannot have more independent directions than its ambient space . . . . . 3

Exercise 2.9.1: Combine two basis vectors and sketch the result . . . . . 3

Exercise 2.9.2: A negative basis coordinate reverses a direction . . . . . 4

Exercise 2.9.3: Recover coordinates from two equations . . . . . 4

Exercise 2.9.4: Coordinates may both be negative . . . . . 4

Exercise 2.9.5: Two coordinates describe a three-entry vector . . . . . 4

Exercise 2.9.6: Verify the remaining coordinate after solving . . . . . 4

Exercise 2.9.7: Read the skew grid and confirm algebraically . . . . . 5

Exercise 2.9.8: A second skew coordinate grid . . . . . 5

Exercise 2.9.9: Rank three, nullity one . . . . . 5
Exercise 2.9.10: Four pivots among five columns . . . . . 5
Exercise 2.9.11: Two pivot directions and three degrees of freedom . . . . . 6
Exercise 2.9.12: A third free-variable calculation . . . . . 6
Exercise 2.9.13: Discard one redundant spanning vector . . . . . 6
Exercise 2.9.14: Select three of five generators . . . . . 6
Exercise 2.9.15: Equal dimensions do not mean equal sets . . . . . 7
Exercise 2.9.16: A three-dimensional column space inside four dimensions . . . . . 7
Exercise 2.9.17: True–false questions about dimension . . . . . 7
Exercise 2.9.18: Uniqueness, nullity, and coordinate maps . . . . . 7
Exercise 2.9.19: Recover rank from nullity . . . . . 8
Exercise 2.9.20: A six-by-eight matrix . . . . . 8
Exercise 2.9.21: The dimension of a homogeneous solution space . . . . . 8
Exercise 2.9.22: Five generators of a four-dimensional span are dependent . . . . . 8
Exercise 2.9.23: Construct prescribed rank and nullity . . . . . 8
Exercise 2.9.24: Construct rank one . . . . . 8
Exercise 2.9.25: Full column-space dimension forces independent columns . . . . . 8
Exercise 2.9.26: A basis need not be the leftmost pivot selection . . . . . 9
Exercise 2.9.27: More vectors than a spanning set must be dependent . . . . . 9
Exercise 2.9.28: Every basis has the same number of vectors . . . . . 9
Exercise 2.9.29: Compute coordinates in a two-dimensional subspace of R4 . . . . . 9
Exercise 2.9.30: Three coordinates in a four-dimensional ambient space . . . . . 9

## 2.9 Dimension and rank

For an ordered basis $\mathcal{B} = (b_1, \ldots, b_p)$, the coordinate vector $[x]_{\mathcal{B}} = (c_1, \ldots, c_p)^T$ means $x = c_1 b_1 + \cdots + c_p b_p$. Coordinates depend on the basis and its order. The dimension of a subspace is the number of vectors in any basis for it. For an $m \times n$ matrix, rank counts pivots, while nullity counts free variables; therefore

$$\dim \operatorname{Col} A = \operatorname{rank} A, \qquad \dim \operatorname{Nul} A = n - \operatorname{rank} A.$$

A subspace's dimension is different from the number of coordinates in its vectors. A two-dimensional subspace of $\mathbb{R}^6$ still consists of six-entry vectors, not two-entry vectors.

### Practice problem 2.9.P1: Count independent directions rather than generators

Put the three vectors into a matrix. Eliminating below column 1 gives

$$\begin{bmatrix} 2 & 3 & -1 \\ -8 & -7 & 6 \\ 6 & -1 & -7 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & -10 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 5 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 2 are pivot columns. Thus a basis for $H$ is $\{(2, -8, 6)^T, (3, -7, -1)^T\}$, and $\dim H = 2$. The third vector is redundant: $v_3 = -11v_1/10 + 2v_2/5$.

### Practice problem 2.9.P2: Turn coordinates into a vector

The coordinate entries are weights, so

$$x = 3(1, .2)^T + 2(.2, 1)^T = (3 + .4, .6 + 2)^T = (3.4, 2.6)^T.$$

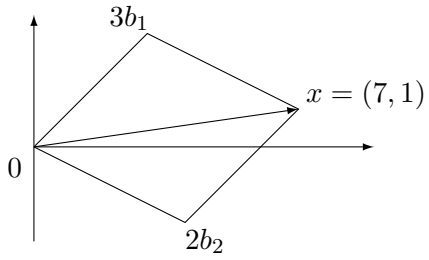
This ordinary coordinate vector differs from $(3, 2)^T$, because the basis is not the standard basis.

### Practice problem 2.9.P3: A subspace cannot have more independent directions than its ambient space

No subspace of $\mathbb{R}^3$ can have dimension four. A four-dimensional subspace would have a basis of four independent vectors in $\mathbb{R}^3$, but any $3 \times 4$ homogeneous system has a free variable, making four such columns dependent. In fact possible dimensions are only $0, 1, 2, 3$; a three-dimensional subspace of $\mathbb{R}^3$ must be $\mathbb{R}^3$ itself.

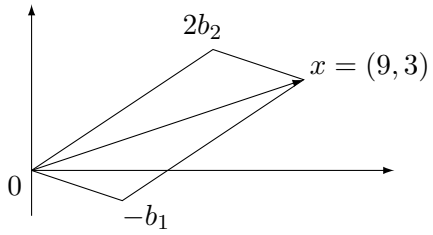
### Exercise 2.9.1: Combine two basis vectors and sketch the result

The supplied coordinate vector gives $x = 3(1, 1)^T + 2(2, -1)^T = (7, 1)^T$. The sketch shows the parallelogram with sides $3b_1 = (3, 3)$ and $2b_2 = (4, -2)$; its diagonal is $x$.




### Exercise 2.9.2: A negative basis coordinate reverses a direction

Now $x = -1(-3, 1)^T + 2(3, 2)^T = (3, -1)^T + (6, 4)^T = (9, 3)^T$. A negative coefficient means travel opposite to $b_1$, not negate the whole answer. The parallelogram below has sides $-b_1$ and $2b_2$.




### Exercise 2.9.3: Recover coordinates from two equations

Write $x = c_1b_1 + c_2b_2$. The two coordinates require $2c_1 - c_2 = 0$ and $-3c_1 + 5c_2 = 7$. The first gives $c_2 = 2c_1$; substituting into the second gives $7c_1 = 7$, hence $c_1 = 1, c_2 = 2$. Thus $\boxed{[x]_{\mathcal{B}} = (1, 2)^T}$, and $(2, -3)^T + 2(-1, 5)^T = (0, 7)^T$ verifies the answer.

### Exercise 2.9.4: Coordinates may both be negative

The equations are $c_1 - 2c_2 = 1$ and $-5c_1 + 3c_2 = 9$. Substitute $c_1 = 1 + 2c_2$ into the second: $-5 - 10c_2 + 3c_2 = 9$, so $c_2 = -2$ and $c_1 = -3$. Therefore $\boxed{[x]_{\mathcal{B}} = (-3, -2)^T}$. Indeed $-3(1, -5)^T - 2(-2, 3)^T = (1, 9)^T$.

### Exercise 2.9.5: Two coordinates describe a three-entry vector

From the first two entries of $c_1b_1 + c_2b_2 = x$, $c_1 - 2c_2 = 2$ and $4c_1 - 7c_2 = 9$. Subtract four times the first equation from the second to get $c_2 = 1$; then $c_1 = 4$. The third entry also matches: $-3(4) + 5(1) = -7$. Thus $\boxed{[x]_{\mathcal{B}} = (4, 1)^T}$. Checking the third coordinate is important: a solution to only two equations would not by itself prove the vector equality in $\mathbb{R}^3$.

### Exercise 2.9.6: Verify the remaining coordinate after solving

The first two equations are $-3c_1 + 7c_2 = 5$ and $2c_1 - 3c_2 = 0$. The second gives $c_1 = 3c_2/2$; the first becomes $(5/2)c_2 = 5$. Thus $c_2 = 2, c_1 = 3$. The last coordinate is $-4(3) + 5(2) = -2$, as required.

Therefore $\boxed{[x]_{\mathcal{B}} = (3,2)^T}$.

### Exercise 2.9.7: Read the skew grid and confirm algebraically

The diagram suggests two units in the $b_1$ direction and minus one unit in the $b_2$ direction for $w$, and one and a half $b_1$ units plus half a $b_2$ unit for $x$. Algebra confirms

$$[w]_{\mathcal{B}} = (2,-1)^T, \qquad [x]_{\mathcal{B}} = (3/2,1/2)^T.$$

Indeed $2(3,0)^T - (-1,2)^T = (7,-2)^T = w$, and $(3/2)(3,0)^T + (1/2)(-1,2)^T = (4,1)^T = x$. These computations turn approximate visual readings into exact answers.

### Exercise 2.9.8: A second skew coordinate grid

Here $c_1 b_1 + c_2 b_2 = (2c_2, 2c_1 + c_2)^T$. Thus for a vector $(u,v)^T$, $c_2 = u/2$ and $c_1 = (v - u/2)/2$. Applying this formula gives

$$[x]_{\mathcal{B}} = (2,-1)^T, \quad [y]_{\mathcal{B}} = (3/2,1)^T, \quad [z]_{\mathcal{B}} = (-1,-1/2)^T.$$

The first matches the picture's movement two $b_1$ steps and minus one $b_2$ step. The requested confirmations are $(3/2)(0,2)^T + (2,1)^T = (2,4)^T = y$ and $-(0,2)^T - (1/2)(2,1)^T = (-1,-5/2)^T = z$.

### Exercise 2.9.9: Rank three, nullity one

The supplied echelon form has pivots in columns 1,3,4, so an original-column basis is

$$\mathcal{C} = \{(1,3,2,5)^T, (2,1,-1,0)^T, (-6,5,9,14)^T\}.$$

The bottom nonzero equation gives $x_4 = 0$; the next gives $x_3 = 0$; the first then gives $x_1 = -3x_2$. Thus $\mathcal{N} = \{(-3,1,0,0)^T\}$ is a null-space basis. Consequently $\dim \operatorname{Col} A = 3$ and $\dim \operatorname{Nul} A = 1$. Their sum is four, the number of columns, not necessarily the number of rows in other examples.

### Exercise 2.9.10: Four pivots among five columns

The pivot positions are columns 1,2,4,5. A column-space basis consists of the columns of

$$C = \begin{bmatrix} 1 & -2 & 5 & 4 \\ 2 & -1 & 5 & 6 \\ -2 & 0 & 1 & -6 \\ 3 & 1 & 1 & 5 \end{bmatrix}.$$

Reading the supplied echelon equations from the bottom upward gives $x_5 = 0$, $x_4 = 0$, $x_2 = -x_3$, and $x_1 = -x_3$. Thus $\{(-1,-1,1,0,0)^T\}$ is a null-space basis. The dimensions are $\boxed{4 \text{ and } 1}$ for column and null spaces, respectively. The basis vectors for the column space are from the original $A$, while the equations for the null space come from its echelon form.

### Exercise 2.9.11: Two pivot directions and three degrees of freedom

The pivots are in columns 1 and 3. Thus a column-space basis is

$$\{(2,3,0,-3)^T,(-5,-8,9,-7)^T\}.$$

The second echelon equation gives $x_3=-x_5$. Substitution in the first gives $x_1=-2x_2-x_4-x_5$. Taking $x_2,x_4,x_5$ as independent parameters gives a null-space basis

$$\{(-2,1,0,0,0)^T,(-1,0,0,1,0)^T,(-1,0,-1,0,1)^T\}.$$

These vectors have independent free-coordinate entries, so they are independent and span all solutions. Therefore $\dim\operatorname{Col}A=2$, $\dim\operatorname{Nul}A=3$.

### Exercise 2.9.12: A third free-variable calculation

The echelon form has pivots in columns 1,2,3. A column-space basis is

$$\{(1,5,4,3)^T,(2,1,6,4)^T,(-4,-9,-9,-5)^T\}.$$

The bottom nonzero equation gives $x_3=x_5$. The preceding equation then gives $x_2=-2x_4-x_5$, and the first gives $x_1=0$. Hence a null-space basis is

$$\{(0,-2,0,1,0)^T,(0,-1,1,0,1)^T\}.$$

The dimensions are $\boxed{3 \text{ and } 2}$. Direct substitution into the original $A$ checks both null vectors, and the free coordinates prove their independence.

### Exercise 2.9.13: Discard one redundant spanning vector

Place the four supplied vectors into $A=[v_1\ v_2\ v_3\ v_4]$. Exact row reduction gives

$$R=\begin{bmatrix}1&-3&0&0\\0&0&1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}.$$

The pivot columns are 1,3,4, so

$$\{(1,-3,2,-4)^T,(2,-1,4,2)^T,(-4,5,-3,7)^T\}$$

is a basis for the supplied span, whose dimension is 3. The omitted vector is $v_2=-3v_1$. The other three are independent by the pivot test; discarding a multiple does not by itself establish that last fact without the reduction.

### Exercise 2.9.14: Select three of five generators

The column matrix of the five vectors reduces exactly to

$$R=\begin{bmatrix}1&0&-2&0&15/2\\0&1&1&0&-7/2\\0&0&0&1&-5/2\\0&0&0&0&0\end{bmatrix}.$$

Thus $v_1, v_2, v_4$ are pivot columns, giving the basis

$$\{(1,-1,-2,3)^T, (2,-3,-1,4)^T, (-1,4,-7,7)^T\}$$

and dimension 3. For a check on spanning, $v_3 = -2v_1 + v_2$ and $v_5 = (15/2)v_1 - (7/2)v_2 - (5/2)v_4$. Thus the two omitted vectors are indeed combinations of the selected ones.

### Exercise 2.9.15: Equal dimensions do not mean equal sets

Four pivot columns in a $4 \times 6$ matrix give four pivot rows, so $\operatorname{Col} A = \mathbb{R}^4$. There are $6 - 4 = 2$ free variables, so $\dim \operatorname{Nul} A = 2$. However $\operatorname{Nul} A \neq \mathbb{R}^2$: its vectors have six entries and it is a subspace of $\mathbb{R}^6$. A basis identifies it with $\mathbb{R}^2$ through coordinates; an identification by coordinates is different from literal equality of the sets.

### Exercise 2.9.16: A three-dimensional column space inside four dimensions

The column space has dimension 3 but consists of four-entry vectors. It is a proper subspace of $\mathbb{R}^4$, not literally $\mathbb{R}^3$. Rank–nullity gives $\dim \operatorname{Nul} A = 7 - 3 = \boxed{4}$. The five words "three-dimensional subspace of $\mathbb{R}^4$" specify both its internal dimension and its ambient space.

### Exercise 2.9.17: True–false questions about dimension

(a) **True.** For an ordered basis, the unique weights in $x = \sum c_i v_i$ are precisely its basis coordinates.

(b) **False.** A line must pass through zero to be a subspace. The line $\{(1, t) : t \in \mathbb{R}\}$ does not contain zero, for instance. A line through zero does form a one-dimensional subspace.

(c) **True.** Pivot columns form a basis for $\operatorname{Col} A$, so their number equals its dimension.

(d) **True.** Pivots plus free variables count all $n$ columns: $\dim \operatorname{Col} A + \dim \operatorname{Nul} A = n$.

(e) **True.** A spanning set of $p$ vectors in a $p$-dimensional space is a basis. If it were dependent, one vector could be deleted without changing its span, leaving at most $p - 1$ generators for a space of dimension $p$, which is impossible.

### Exercise 2.9.18: Uniqueness, nullity, and coordinate maps

(a) **True.** Two coefficient representations would subtract to a dependence relation among basis vectors; independence forces the corresponding coefficients to be identical.

(b) **False.** The null-space dimension is the number of *free* variables, not all variables. For $A = I_n$ it is zero despite there being $n$ variables.

(c) **True.** Rank is the number of pivots, which is the column-space dimension.

(d) **True.** The coordinate map is a linear bijection to $\mathbb{R}^p$, with inverse $(c_1, \ldots, c_p)^T \mapsto \sum c_i v_i$. Thus it preserves addition and scalar multiplication and assigns one coordinate vector to every vector of $H$.

(e) **True.** If $p$ independent vectors in a $p$-dimensional space did not span it, a vector outside their span could be appended to produce $p + 1$ independent vectors, contradicting the dimension.

### Exercise 2.9.19: Recover rank from nullity

The null space has a basis of three vectors, so its dimension is 3. There are seven columns. Thus $\operatorname{rank} A = 7 - 3 = \boxed{4}$ by rank–nullity. The five rows impose the separate upper bound $\operatorname{rank} A \leq 5$, which this answer respects.

### Exercise 2.9.20: A six-by-eight matrix

There are eight columns and nullity three, so $\operatorname{rank} A = 8 - 3 = \boxed{5}$. This is possible because five pivots fit into six rows.

### Exercise 2.9.21: The dimension of a homogeneous solution space

The solution space is $\operatorname{Nul} A$. Rank–nullity with eight columns and rank seven gives $\dim \operatorname{Nul} A = 8 - 7 = \boxed{1}$. Its vectors have eight entries even though only one parameter is needed to describe them.

### Exercise 2.9.22: Five generators of a four-dimensional span are dependent

If the five vectors were independent, they would be a basis for their own span, making its dimension five by the definition of dimension. The given dimension is four, a contradiction. Hence the vectors are dependent. The ambient value of $n$ does not change this argument.

### Exercise 2.9.23: Construct prescribed rank and nullity

Take

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Its column-space basis is $\{e_1, e_2\} \subset \mathbb{R}^3$, so its rank is 2. Its homogeneous solutions are $(0, 0, s, t, u)^T$, so nullity is 3. Thus the requested $3 \times 5$ shape and both dimensions are achieved.

### Exercise 2.9.24: Construct rank one

Choose

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Every column is a multiple of the nonzero first column, so the column space is exactly one line. Hence $A$ has rank 1; it has the required $3 \times 4$ shape.

### Exercise 2.9.25: Full column-space dimension forces independent columns

The $p$ columns span $\operatorname{Col} A$ by definition, and this space has dimension $p$ by hypothesis. A spanning set with exactly its dimension many vectors is a basis. Therefore the columns are independent. Equivalently rank $p$ in a matrix with $p$ columns leaves no free variables.

### Exercise 2.9.26: A basis need not be the leftmost pivot selection

The five named columns lie in $\operatorname{Col} A$, which has dimension five because $\operatorname{rank} A = 5$. They are independent by hypothesis. An independent set of five vectors in a five-dimensional space is a basis, so these columns span $\operatorname{Col} A$. The pivot-column algorithm is one way to find a basis, not a claim that all bases must use the same indices.

### Exercise 2.9.27: More vectors than a spanning set must be dependent

(a) Since $a_j \in W = \operatorname{Span}\{b_1, \ldots, b_p\}$, there are coefficients $c_{1j}, \ldots, c_{pj}$ with $a_j = \sum_i c_{ij} b_i$. Collect them in $c_j \in \mathbb{R}^p$ to obtain $a_j = Bc_j$.

(b) The matrix $C = [c_1 \ \cdots \ c_q]$ has $p$ rows and $q > p$ columns. Its homogeneous system has more variables than possible pivots, so at least one variable is free. Thus some $u \neq 0$ satisfies $Cu = 0$.

(c) The column identities in part (a) give $A = BC$. Therefore $Au = B(Cu) = B0 = 0$. Since $u \neq 0$, its entries furnish a nontrivial linear dependence among $a_1, \ldots, a_q$.

This proves the claim without assuming the two supplied lists were bases.

### Exercise 2.9.28: Every basis has the same number of vectors

Let $\mathcal{A}, \mathcal{B}$ be bases of $W$, with $q, p$ vectors, respectively. If $q > p$, then Exercise 27 applied to the spanning set $\mathcal{B}$ would make $\mathcal{A}$ dependent, contrary to its being a basis. Hence $q \leq p$. Interchanging the roles gives $p \leq q$. Thus $p = q$. This justifies defining dimension using *any* basis.

### Exercise 2.9.29: Compute coordinates in a two-dimensional subspace of R4

The vectors $v_1, v_2$ are independent: their first two coordinates have $2 \times 2$ determinant $15(-10) - 14(-5) = -80 \neq 0$. Thus they are a basis for their span $H$. A direct calculation gives

$$2v_1 - v_2 = (30 - 14, -10 + 10, 24 - 13, 14 - 17)^T = (16, 0, 11, -3)^T = x.$$

Therefore $x \in H$ and $\boxed{[x]_{\mathcal{B}} = (2, -1)^T}$. These exact integer products also verify the computer-assisted answer.

### Exercise 2.9.30: Three coordinates in a four-dimensional ambient space

Put $V = [v_1 \ v_2 \ v_3]$ and row reduce the augmented matrix. Exact arithmetic gives

$$[V \mid x] \ \sim \ \left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The three coefficient pivots show that $v_1, v_2, v_3$ are independent, so $\mathcal{B}$ is a basis for $H$. The consistent right block gives $[x]_{\mathcal{B}} = \boxed{(-2, 1, 1)^T}$. For an independent check,

$$-2v_1 + v_2 + v_3 = (12 + 8 - 9, -6 + 0 + 4, 18 + 7 - 8, -8 - 3 + 3)^T = (11, -2, 17, -8)^T = x.$$

Thus the coordinate vector has three entries while $x$ itself has four.